HarvestMetric: leakage-safe evaluation of crop-yield models

Abir Ahmmed · Technical report · November 2026 · github.com/devbyabir/harvestmetric

Abstract. An earlier crop-yield notebook reported $R^2 \approx 1.00$. I show that the score came from duplicated records leaking across a row-level train/test split of a synthetic dataset. With farms kept apart, the same model scores $R^2 = -0.21$, and no tuned model beats the mean (permutation p = 0.55). I then pose a defensible forecasting question on real public data: Iowa county corn-for-grain yield, forecast on 1 August from the county's past yields and season-to-date weather. On untouched test years 2012–2025, gradient boosting on trend deviations lowered RMSE from 21.8 to 17.6 bu/acre against a county-trend baseline. However, the gain's year-resampled 95% interval reaches zero ([-0.00, +9.46]), and in training-period cross-validation the model did not beat the baseline. By a rule fixed before testing, the result is reported as promising but not established, and no live forecasts are published.

1 Research questions

Q1 (Part A): Was the original $R^2 \approx 1.00$ evidence of forecasting ability? **Q2** (Part B): On real observational data, does season-to-date weather improve a 1 August forecast of county corn yield beyond simple baselines built from the county's own yield history? Does any improvement hold across years and across regions?

2 Data and licensing

Part	Source	Terms	In repository
A	Smart Farming Sensor Data for Yield Prediction (A. Soundankar, Kaggle)	Apache-2.0; synthetic	Not redistributed; aggregates only
B	USDA NASS Quick Stats: Iowa county corn-for-grain yield, bu/acre, 1926–2025	CC0 1.0	Processed panel committed
B	NASA POWER daily point API v2.10.3: T2M_MAX, T2M_MIN, PRECTOTCORR, 1981–2026	Citation required	Features in panel
B	US Census Bureau 2024 Gazetteer: county internal points	US Gov. work (17 U.S.C. §105)	Coordinates in panel

The original working copy (5,000 rows) was verified record-identical to the official Kaggle file (500 rows): dates were reformatted and every record repeated 10 times. Retrieval dates, query parameters and SHA-256 checksums are in *data/processed/manifest.json*. Raw downloads are re-created by *python -m harvestmetric.real.acquire*.

3 Part A: the leakage case study

With each record present ten times, a row-level *KFold* places near-certain copies of every test row in training, and flexible models memorise them. I removed exact duplicates, held out 100 farms (20%, stratified by crop) once, and tuned five models with nested, farm-grouped CV on the remaining 400. The forecast point is the sowing date. Season totals, in-season sensor readings, harvest date and growing duration are excluded. A permutation test and bootstrap intervals judge skill. An audit found the data's coordinates contradict its region labels and its columns behave as independent uniform draws. The publisher describes it as simulated.

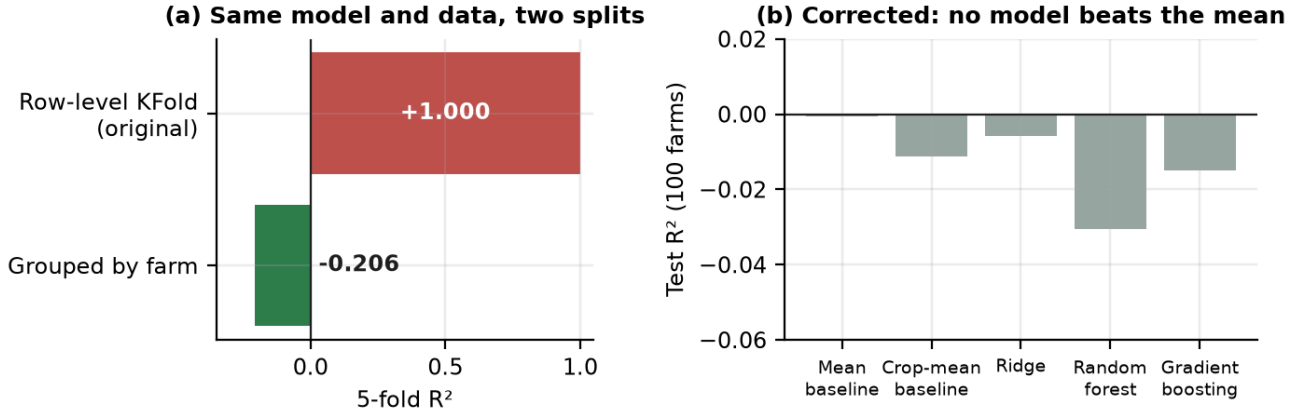


Figure 1. (a) The original split's R^2 is an artefact of duplication. (b) Under a farm-grouped hold-out, test R^2 is ≈ 0 for every model.

Model (at sowing)	CV R^2	Test R^2	Test MAE kg/ha
Mean baseline	-0.010	-0.001	1,013
Crop-mean baseline	-0.017	-0.011	1,011
Ridge	-0.018	-0.006	1,013
Random forest	-0.063	-0.031	1,021
Gradient boosting	-0.057	-0.015	1,008

Answer to Q1: no. Test R^2 for the CV-selected model is -0.006 (95% CI [-0.060, +0.001]). Predictions collapse to the mean, so low- and high-yield farms are missed by about 1,500 kg/ha. Valid evaluation shows the data contains no learnable yield signal, so no prediction feature is offered.

4 Part B: forecasting task

Unit: Iowa county × year (99 counties). **Target:** NASS final corn-for-grain yield (bu/acre; 1 bu/ac ≈ 0.0627 t/ha). **Forecast date:** 1 August of year t. **Information allowed:** the county's published yields for t−30…t−1, and NASA POWER weather from 1 April to 31 July of year t at the county's Census internal point. **Features:** a county linear trend fitted to the previous 30 years, last year's deviation from that trend (flagged when unpublished), May–July growing degree-days (10/30 °C), June–July degree-days above 30 °C, April–May, June and July precipitation, and mean July maximum temperature. **Leakage controls**, enforced in code and tests: weather after 31 July cannot enter a feature, history uses only years before t, no feature uses another county's same-year yield, and identifiers are never inputs. Combined "other counties" rows and withheld values are dropped. Counties not published after 2014 are left unscored rather than imputed.

5 Validation design and models

- **Chronological test:** train 1981–2011 (3,069 county-years); untouched test 2012–2025 (1,282). The test years were scored once, after model selection.
- **Tuning:** expanding-window CV inside the training years (validation blocks 1997–99, …, 2009–11, each fitted only on earlier years). Selection used CV RMSE.
- **Baselines:** the county's 30-year mean, its previous-year yield, and its 30-year linear trend.
- **Models:** Ridge and histogram gradient boosting, both predicting the deviation from the county trend. Ridge captures linear weather effects; boosting allows heat and drought thresholds.
- **Uncertainty:** split-conformal 80%/90% intervals calibrated on the time-ordered CV residuals. Gains are tested with a bootstrap that resamples whole *years*, because one season's weather shock hits all counties at once.
- **Pre-registered gate:** forecasts are published only if the model beats the best baseline in CV *and* its test RMSE gain over the strongest baseline has a 95% interval above zero.
- **Geographic check (separate):** leave one of NASS's nine crop-reporting districts out, within 1981–2011.

6 Results

Model	CV RMSE	Test MAE	Test RMSE	Test R^2
Historical mean	39.7	38.2	41.6	-1.31
Previous year	19.4	19.7	25.4	+0.14
County trend	18.8	16.5	21.8	+0.37
Trend + Ridge	23.4	14.2	18.9	+0.52
Trend + gradient boosting	19.5	13.5	17.6	+0.59

bu/acre; test = 2012–2025. Selected by CV: Trend + gradient boosting.

Gain of selected model over	RMSE gain	95% CI (years)
Historical mean	+24.03	[+16.52, +30.92]
Previous year	+7.77	[+4.33, +11.32]
County trend	+4.20	[-0.00, +9.46]

Positive = selected model better. Rolling annual refit: RMSE 17.9 vs 21.8, CI [-0.39, +9.34].

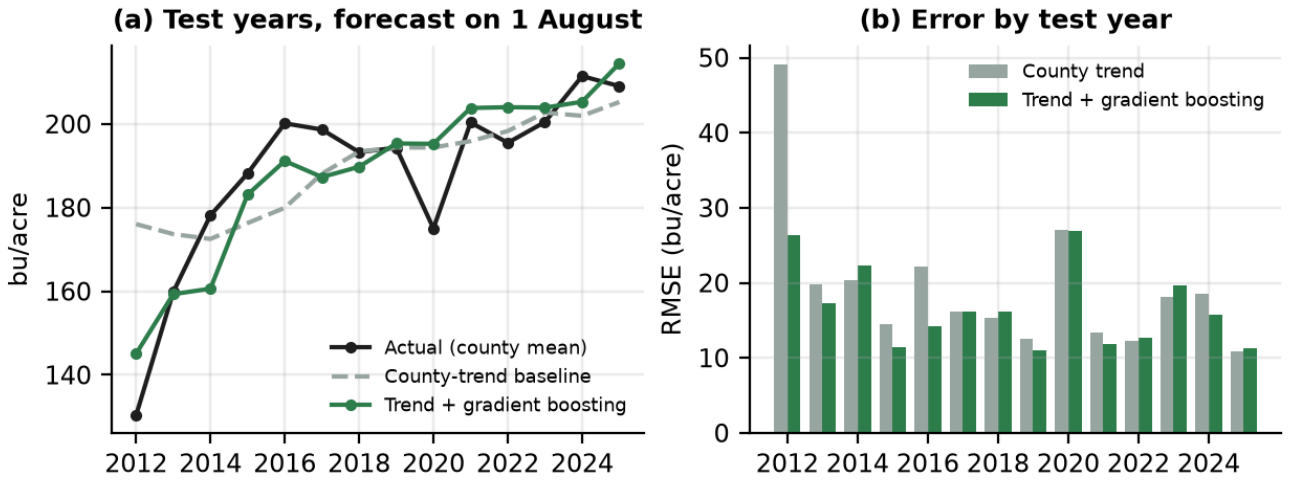


Figure 2. (a) County-average actual yield and 1 August forecasts. (b) RMSE by test year: most of the gain comes from 2012.

The model clearly beats the historical-mean and previous-year baselines. Its gain over the county trend (+4.2 bu/acre, 19% lower RMSE) is positive in 97.5% of year resamples, but the interval reaches zero. In CV the model was *worse* than the trend (19.5 vs 18.8). The early folds learned from only three disaster years (1983, 1988, 1993) and over-predicted heat losses in 1997, 1999 and 2002, when yields were above trend despite high heat. The weather–yield relationship is not stable across eras, so the gate is **not passed**. Conformal intervals covered 86.2% (80%) and 94.1% (90%) of test county-years overall (half-widths ±25 and ±33 bu/acre), but only 69% and 68% at 80% in 2012 and 2020. Exchangeability fails within a shock year. In the geographic check the model beat the trend in 9 of 9 held-out districts (RMSE 16.8 vs 22.8). The same seasons appear on both sides there, so this shows spatial transfer, not skill on new seasons.

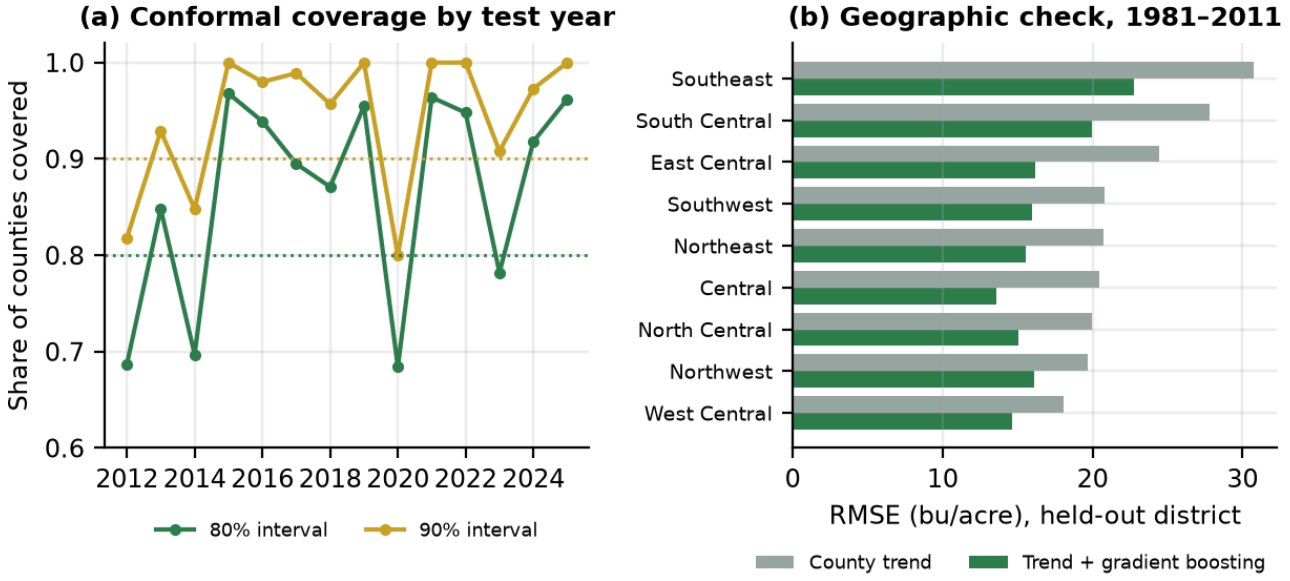


Figure 3. (a) Interval coverage collapses in shock years (2012 drought, 2020 derecho). (b) Leave-one-district-out RMSE.

7 Error analysis

- **Extreme heat** (June–July degree-days above the training 90th percentile; defined from weather only): model RMSE 22.2 vs trend 36.5; bias +7.5 vs +22.3. Weather helps most where the trend fails most.
- **Large shortfalls** (yield >15% below trend; outcome-defined, descriptive only, n = 141): still under-forecast severity, bias +24.4 bu/acre (trend +46.4).
- **2020:** the 10 August derecho damaged central-Iowa corn after the forecast date. No 1 August information could anticipate it (bias +20.2 bu/acre).
- Permutation importance and SHAP-style explanations were **not** produced, because the model did not pass validation. Associations in this data are not causal effects.

8 Limitations

Weather is taken at a single point per county from a ~0.5° × 0.625° grid. Neighbouring counties often share a cell, and the point ignores where corn is grown. Targets are NASS's current estimates, which can differ from values published in real time. Unpublished counties after 2014 are missing not at random. Fourteen test years give wide intervals. The model set was deliberately small, and richer inputs (soil, planting dates, satellite vegetation indices) were out of scope. Part A uses synthetic data and says nothing about real crops.

9 Conclusion

The original $R^2 \approx 1.00$ was leakage. On real data, a carefully time-separated evaluation finds that season-to-date weather *probably* improves 1 August county yield forecasts in drought years, but the evidence does not meet a pre-specified standard: CV across 1997–2011 disagrees, and the test gain is not reliably above zero. The contribution is a reproducible, leakage-tested pipeline that reports this honestly, rather than a headline accuracy number.

Authorship note. The original exploratory notebook was developed during an internship together with a colleague and is not included. This rebuild, data pipeline, evaluation and report were produced by the author with the assistance of an AI coding assistant.

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